Lecture 2

Compactness, connectedness and gluing

Objectives. *Open sets let us define properties that distinguish spaces. Compactness expresses a finiteness condition on open covers. Connectedness rules out a separation into two open pieces. We prove how continuous maps preserve these properties, then use them to identify spaces formed by gluing. The main criterion turns a continuous bijection into a homeomorphism when its domain is compact and its target is Hausdorff.*

By the end of this lecture you will be able to:

- Prove the compact-to-Hausdorff criterion for a homeomorphism.
- Distinguish spaces by compactness, and by connectedness after removing a point.
- Test continuity out of a quotient and identify its points correctly.
- Describe the torus, Möbius band, projective plane and rotation group as quotients.

2.1 Compactness

Definition 2.1 (Compactness). An *open cover* of a space X is a family of open sets $(U_\alpha)_{\alpha\in I}$ whose union is X. A *subcover* is a subfamily that still covers X. The space is *compact* if every open cover has a finite subcover.

A subset $K \subseteq X$ is called compact when its subspace topology is compact. Equivalently, every cover of K by open sets of X has a finite subcover. This follows because the open sets of K are intersections $K \cap U$ with ambient open sets U (§ 1.4).

The figure illustrates the condition on an interval. A finite selection must cover every point, including both endpoints.

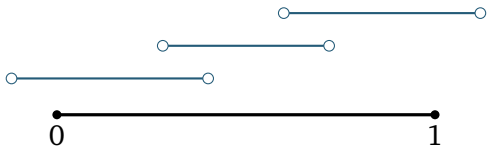


Figure 2.1: The three open intervals shown cover $[0, 1]$. For compactness, a finite subcover must exist for every open cover, not just this one.

Every finite space is compact. Choose one cover member containing each point. The resulting selection is finite. The empty space needs no members. In contrast, the real line is not compact. The open cover $\{(-n, n) : n \geq 1\}$ has no finite subcover, since a finite selection is contained in its member with the largest index.

Theorem 2.2 (Continuous images and closed subsets). *A continuous image of a compact space is compact. A closed subset of a compact space is compact.*

Proof. Let $f\colon X \to Y$ be continuous with X compact. Take an ambient open cover (V_α) of $f(X)$. Its inverse images cover X and are open. Choose finitely many of them covering X. The corresponding sets V_α cover $f(X)$.

Now let $C \subseteq X$ be closed, with X compact. Take an ambient open cover of C and add the open set $X \setminus C$. This covers X and therefore has a finite subcover. Discarding $X \setminus C$, if selected, leaves finitely many members covering C. ∎

We use two results about metric spaces with their proofs cited. *Heine–Borel* says that a subset of $\mathbb{R}^n$ is compact exactly when it is closed and bounded (Munkres, Theorem 27.3). Bounded means contained in some ball of finite radius. Thus closed intervals, closed balls and unit spheres are compact. The hypotheses refer to Euclidean space. Closed and bounded need not imply compact in a general metric space.

A metric space is *sequentially compact* if every sequence in it has a subsequence converging to a point of that space. This is the notion used in *Linear Algebra for Physics*, Lecture 13. It is equivalent to open-cover compactness for metric spaces (Munkres, Theorem 28.2). We use the open-cover definition on spaces without a specified metric.

The circle is compact by Heine–Borel, whereas the real line is not. They cannot be homeomorphic. A homeomorphism would take the compact circle onto the line. This settles one distinction promised in Lecture 1.

2.2 Compact spaces and Hausdorff targets

Hausdorffness means that distinct points have disjoint open neighbourhoods (§ 1.5). It supplies the other hypothesis in the homeomorphism criterion.

Proposition 2.3 (Compact subsets are closed). *A compact subset of a Hausdorff space is closed.*

Proof. Let $K \subseteq Y$ be compact and $y \in Y \setminus K$. For each $x \in K$, choose disjoint open neighbourhoods U_x of x and V_x of y. Finitely many sets $U_{x_1}, \dots, U_{x_m}$ cover K. The open neighbourhood $V_{x_1} \cap \cdots \cap V_{x_m}$ of y misses their union, so misses K. Thus $Y \setminus K$ is open. If K is empty, it is closed directly. ∎

Continuity can also be tested on closed sets. Inverse images commute with complements, so the open-set test is equivalent to the closed-set test.

Corollary 2.4 (Compact-to-Hausdorff criterion). *A continuous bijection from a compact space to a Hausdorff space is a homeomorphism.*

Proof. Let $f\colon X \to Y$ satisfy the hypotheses. For a closed subset $C \subseteq X$, compactness of X makes C compact. Its image $f(C)$ is compact, hence closed in Y. This image is the inverse image of C under $f^{-1}\colon Y \to X$. The closed-set test proves continuity of the inverse. ∎

Lecture 1's continuous bijection $[0, 2\pi) \to S^1$ does not meet the criterion. Its domain is not compact, since it is not closed in $\mathbb{R}$. We will instead use $[0, 1]$ and identify its two endpoints.

2.3 Connectedness

Definition 2.5 (Connectedness and paths). A *separation* of a space X is a pair of disjoint, nonempty open sets whose union is X. The space is *connected* if it has no separation. A *path* from x to y in X is a continuous map $\gamma\colon [0,1] \to X$ with $\gamma(0) = x$ and $\gamma(1) = y$. The space is *path-connected* if such a path exists for every pair $x, y \in X$.

A subset is connected or path-connected when its subspace topology has the corresponding property.

Theorem 2.6 (Intervals, images and paths). *The interval $[0,1]$ is connected. Continuous images preserve connectedness and path-connectedness. Every path-connected space is connected.*

Proof. Suppose $[0,1] = U \cup V$ is a separation. Choose $a \in U$ and $b \in V$, with $a < b$ after exchanging the names if necessary. Put $c = \sup(U \cap [a,b])$. If $c \in U$, then $c < b$, and openness of U gives a point of $U \cap [a,b]$ larger than c. If $c \in V$, then $c > a$, and openness of V forbids points of U immediately below c. This contradicts the defining property of the supremum. Both cases are impossible, so no separation exists.

For a continuous map $f\colon X \to Y$, a separation of $f(X)$ pulls back to a separation of X. Thus a connected domain has connected image. If X is path-connected, composing its paths with f gives paths between any two image points.

Finally, a path-connected space cannot have a separation $U \cup V$. A path joining a point of U to a point of V would have connected image, since $[0,1]$ is connected. Its intersections with U and V would separate that image. ∎

For example, a square and a ball are path-connected. The segment $\gamma(t) = (1-t)x + ty$ stays in the set and joins any two of its points.

A *connected component* is a connected subset maximal under inclusion. The component of $x \in X$ is the union of all connected subsets containing x. This union is connected. In any proposed separation, each of its connected subsets must lie wholly on the side containing x. It is maximal by its definition. Two components that meet therefore coincide, so components partition X.

Removing a point distinguishes a line from a plane, as the figure shows.

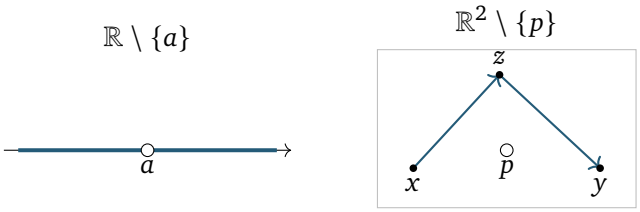


Figure 2.2: Removing a point separates a line into two open rays. In a plane, a broken line can join two points while avoiding the removed point.

The space $\mathbb{R} \setminus \{a\}$ has a separation into its two open rays. In $\mathbb{R}^n \setminus \{p\}$, with $n \geq 2$, take points x and y and choose z outside the two lines through p, x and through p, y. The segments from x to z and from z to y avoid p. They give the path

$$\gamma(t) = \begin{cases} (1-2t)x + 2tz, & 0 \leq t \leq \frac{1}{2}, \\ (2-2t)z + (2t-1)y, & \frac{1}{2} \leq t \leq 1. \end{cases}$$

The two formulas agree at $t = 1/2$. Each affine formula is continuous, so the preimage of a closed set is closed on its half-interval. It is also closed in $[0,1]$, since both half-intervals are closed there. Their finite union is closed, proving continuity of γ. Thus the punctured space is path-connected.

A homeomorphism restricts to a homeomorphism after deleting a point and its image. The punctured line and plane have different connectedness properties, so $\mathbb{R} \not\cong \mathbb{R}^2$. The radial map $x \mapsto x/\|x\|$ also shows that S^n is path-connected for $n \geq 1$, as an image of $\mathbb{R}^{n+1} \setminus \{0\}$.

The converse to path-connected implies connected is false. A cited example is the closure of $\{(x, \sin(1/x)) : 0 < x \leq 1\}$ in $\mathbb{R}^2$, the *topologist's sine curve* (Munkres, §24, Example 7). Its analysis is reserved for Sheet 2.

2.4 Quotients and orbit spaces

Definition 2.7 (Quotient topology)**.** For an equivalence relation $\sim$ on X, the *quotient set* $X/\!\sim$ consists of the equivalence classes. The projection $q\colon X \to X/\!\sim$ sends each point to its class. The *quotient topology* is defined by

$$V \subseteq X/\!\sim \text{ is open} \quad \Longleftrightarrow \quad q^{-1}(V) \text{ is open in } X.$$

Inverse images preserve unions and finite intersections, so this is a topology. The projection is continuous and surjective. Hence a quotient of a compact or path-connected space retains that property.

Proposition 2.8 (Continuity out of a quotient)**.** *A map $\bar{f}\colon X/\!\sim \to Y$ is continuous exactly when $\bar{f} \circ q$ is continuous. A map $f\colon X \to Y$ that is constant on equivalence classes induces a unique map $\bar{f}$ with $f = \bar{f} \circ q$.*

Proof. Composition with continuous q preserves continuity. Conversely, if $\bar{f} \circ q$ is continuous, then for open $W \subseteq Y$ the set $q^{-1}(\bar{f}^{-1}(W))$ is open. The quotient definition says that $\bar{f}^{-1}(W)$ is open. The induced map is given by $\bar{f}([x]) = f(x)$, which is well-defined precisely when f is constant on each class. Surjectivity of q makes it unique. ∎

Define the circle map $c\colon [0,1] \to S^1$ by

$$c(t) = (\cos(2\pi t), \sin(2\pi t)).$$

The figure depicts its endpoint identification.

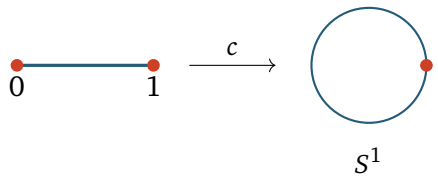


Figure 2.3: The circle map c sends the paired endpoints to the same point. Its induced map from the quotient will be a homeomorphism.

This map is continuous and surjective onto S^1. Its only repeated value on $[0,1]$ is the common endpoint value. The induced map $[0,1]/(0 \sim 1) \to S^1$ is therefore a continuous bijection. Its domain is compact and its target Hausdorff, so it is a homeomorphism by

Corollary 2.4. A *fibre* is the inverse image of one point. This gives a general method: construct a continuous map with exactly the desired classes as its fibres, then apply the criterion.

Recall that a group has an associative product, an identity e and inverses. An *action by homeomorphisms* of a group G on X assigns a homeomorphism $x \mapsto gx$ to each $g \in G$, with $ex = x$ and $(gh)x = g(hx)$. The *orbit* of x is $\{gx : g \in G\}$. The *orbit space* X/G has the quotient topology for these orbits.

One useful case lets us prove Hausdorffness explicitly. An *isometry* preserves a metric. Suppose $a\colon X \to X$ is an isometry with $a^2 = \mathrm{id}$. Its orbit classes are $[x] = \{x, ax\}$, with duplicates omitted. The orbit distance is

$$d_{\mathrm{orb}}([x],[y]) = \min\{d(x,y), d(x,ay)\}.$$

The isometry and $a^2 = \mathrm{id}$ give $d(ax,y) = d(a^2x,ay) = d(x,ay)$. Thus changing either representative interchanges the two distances. The minimum vanishes exactly when the orbits agree. Symmetry follows from the isometry. For the triangle inequality, choose maps $g, h \in \{\mathrm{id}, a\}$ realizing the two relevant minima. Then

$$d(x,ghz) \le d(x,gy) + d(gy,ghz) = d(x,gy) + d(y,hz).$$

This bounds the orbit distance from $[x]$ to $[z]$ by the sum of the other two. Thus the formula defines a metric. Its balls are $q(B_r(x))$, and their inverse images are $B_r(x) \cup B_r(ax)$. Thus orbit-metric balls are quotient-open. Conversely, the preimage of a quotient-open set containing $[x]$ contains some $B_r(x)$. Its image is an orbit-metric ball inside that set, so the set is metric-open. The topologies agree. The orbit space is therefore Hausdorff by Proposition 1.11.

For $n \ge 1$, the *real projective space* $\mathbb{R}\mathrm{P}^n$ is $S^n/(x \sim -x)$. Its points are the lines through the origin in $\mathbb{R}^{n+1}$. The antipodal map is an isometry, so $\mathbb{R}\mathrm{P}^n$ is Hausdorff. It is compact and path-connected as an image of S^n. Writing $D^n = \{v \in \mathbb{R}^n : \|v\| \le 1\}$, the map

$$D^n \longrightarrow \mathbb{R}\mathrm{P}^n, \qquad v \longmapsto [(v, \sqrt{1 - \|v\|^2})]$$

is continuous and surjective. The upper hemisphere consists of unit vectors whose last coordinate is nonnegative. Every line has a representative there. That representative is unique unless its last coordinate vanishes, when there are two antipodal representatives. Thus this realizes $\mathbb{R}\mathrm{P}^n$ as D^n with antipodal points of its boundary sphere identified, again by the compact-to-Hausdorff criterion.

The Hausdorff conclusion fails for general orbit spaces. Let the additive group $\mathbb{Q}$ act on $\mathbb{R}$ by translation. A nonempty open union U of orbits contains an interval. For any $x \in \mathbb{R}$, density of the rationals supplies $r \in \mathbb{Q}$ with $x - r$ in that interval. Invariance under translation by r gives $x \in U$. Hence the only open sets of $\mathbb{R}/\mathbb{Q}$ are the empty set and the whole space. There are distinct classes, for example those of 0 and $\sqrt{2}$, so this quotient is not Hausdorff. The Hausdorff condition will matter for the spaces constructed in Lectures 7 and 8.

2.5 Rotations as a glued ball

The rotation group is the matrix set

$$\mathrm{SO}(3) = \{R \in \mathbb{R}^{3\times 3} : R^T R = \mathbb{1},\ \det R = 1\},$$

with its Euclidean subspace topology. A vector v with $\|v\| \le \pi$ will specify an axis and an angle. The figure shows the identification that occurs at angle π, a half-turn. The vector n marks a unit axis.

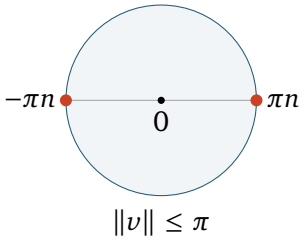


Figure 2.4: A cross-section of the closed rotation ball. Interior vectors are distinct rotations. The matching red boundary points πn and $-\pi n$ represent the same half-turn.

Proposition 2.9 (Axis–angle ball). *The closed ball* $\{v \in \mathbb{R}^3 : \|v\| \le \pi\}$ *with antipodal boundary points identified is homeomorphic to* $\mathrm{SO}(3)$. *Consequently,* $\mathrm{SO}(3) \cong \mathbb{R}\mathrm{P}^3$.

Proof. Define $A_v(w) = v \times w$ for $w \in \mathbb{R}^3$. Put $r = \|v\|$ and define

$$R(v) = \mathbb{1} + \frac{\sin r}{r} A_v + \frac{1 - \cos r}{r^2} A_v^2.$$

At $r = 0$, the coefficients have limiting values 1 and $1/2$. The map is therefore continuous there and $R(0) = \mathbb{1}$. For $v = rn$ with $\|n\| = 1$, the vector identity $n \times (n \times w) = n(n \cdot w) - w$ gives

$$R(v)w = (\cos r)w + (1 - \cos r)n(n \cdot w) + (\sin r)n \times w.$$

This fixes n and rotates its perpendicular plane through angle r. Thus it is orthogonal with determinant 1. At $r = \pi$ it equals $2nn^T - \mathbb{1}$, unchanged by $n \mapsto -n$.

Conversely, every $R \in \mathrm{SO}(3)$ has a unit fixed vector n. Indeed, orthogonality forces all eigenvalues to have modulus 1. Its nonreal eigenvalues occur in conjugate pairs of product 1. Its real eigenvalues are ± 1. After removing the conjugate pairs, an odd number of real eigenvalues remains. Their product is 1, so an even number are -1. At least one is therefore $+1$.

The perpendicular plane is invariant. Since $Rn = n$ and R is orthogonal, for $w \perp n$ we have $Rw \cdot n = Rw \cdot Rn = w \cdot n = 0$. The restriction has determinant 1, so is a plane rotation. Choosing its angle in $[-\pi, \pi]$ represents R as $R(\theta n)$.

There are no further identifications. The trace is $1 + 2\cos r$, so it determines $r \in [0, \pi]$. For $0 < r < \pi$, the difference $R(v) - R(v)^T = 2 \sin r \, A_n$ determines n. At $r = \pi$, the matrix $2nn^T - \mathbb{1}$ determines the line spanned by n, hence just the pair $\{n, -n\}$. At $r = 0$ there is only the vector 0.

We have a continuous bijection from the compact quotient ball to the Hausdorff matrix space $\mathrm{SO}(3)$. It is a homeomorphism. Rescaling the ball to D^3 and using the hemisphere construction above identifies it with $\mathbb{R}\mathrm{P}^3$. ∎

Lecture 4 will use this model to distinguish loops representing full rotations. No property of loops is needed for the identification here.

2.6 Three quotients of a square

We determine compactness, path-connectedness and Hausdorffness for the spaces obtained by the edge pairings in the figure. All start from $Q_0 = [0,1]^2$. Matching arrows indicate which points are identified. Apply also all identifications forced transitively by these pairings.

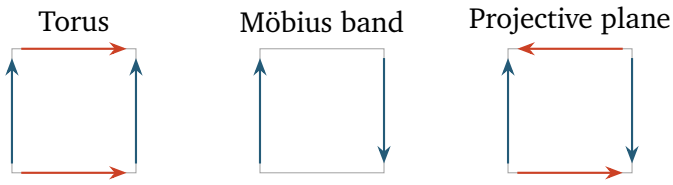


Figure 2.5: The torus pairs opposite edges without reversal. The Möbius band pairs only the vertical edges, with reversal. The projective plane pairs antipodal points of the square boundary.

Each quotient is compact and path-connected, since the square has both properties and the projection is continuous. Hausdorffness needs a separate argument. It need not survive a quotient, as $\mathbb{R}/\mathbb{Q}$ showed.

For the *torus*, the pairings are

$$(0,t) \sim (1,t) \quad \text{and} \quad (s,0) \sim (s,1).$$

Use the circle map c from § 2.4. The product $T^2 = S^1 \times S^1 \subseteq \mathbb{R}^4$ is the torus. The map $Q_0 \to T^2$ given by $(s,t) \mapsto (c(s), c(t))$ is continuous and onto.

Equality in the product is coordinatewise. In each coordinate, c identifies only the two endpoints. Thus the map's fibres are exactly the stated equivalence classes, including the single class of all four corners. The induced continuous bijection is a homeomorphism, because the quotient is compact and T^2 is Hausdorff.

For the *Möbius band*, the pairing is $(0,t) \sim (1, 1-t)$.

Identify S^1 with unit complex numbers. On the cylinder $S^1 \times [-1,1] \subseteq \mathbb{R}^3$, the map $(z,u) \mapsto (-z,-u)$ is an isometry whose square is the identity. Its orbit space is therefore Hausdorff by the orbit metric above. The map

$$Q_0 \longrightarrow (S^1 \times [-1,1])/\{\pm 1\}, \qquad (s,t) \longmapsto [(e^{i\pi s}, 2t-1)]$$

is continuous and onto. Every circle point or its negative lies on the upper semicircle.

For $0 < s < 1$ that representative is unique. At the two ends, changing 1 to -1 reverses the other coordinate, giving precisely the stated edge pairing. The induced bijection is a homeomorphism by the same compact-to-Hausdorff criterion.

For the *projective plane*, the pairings are

$$(0,t) \sim (1, 1-t) \quad \text{and} \quad (s,0) \sim (1-s, 1).$$

Put $w = (2s-1, 2t-1)$. The centred square maps homeomorphically to D^2 by radial rescaling:

$$w \longmapsto \frac{\|w\|_\infty}{\|w\|} w \quad (w \neq 0), \qquad 0 \longmapsto 0,$$

where $\|w\|_\infty$ is the maximum absolute coordinate and $\|w\|$ is Euclidean. The inverse sends nonzero u to $\|u\|\ u/\|u\|_\infty$. Their norms tend to zero at the origin, so both maps are continuous there as well. The square boundary maps to the unit circle and antipodes map to antipodes.

The hemisphere construction therefore identifies this square quotient with $\mathbb{R}P^2$, whose Hausdorffness was proved using its orbit metric.

The results, including the rotation ball, are as follows.

Quotient	Compact and path-connected	Hausdorff witness
Torus	yes	subspace of $\mathbb{R}^4$
Möbius band	yes	isometric orbit space
Projective plane	yes	antipodal orbit metric
Rotation ball	yes	$SO(3) \subseteq \mathbb{R}^9$

Self-check. Prove that S^1 and $[0,1]$ are not homeomorphic by removing one point. Explain why deleting an endpoint of the interval is not a useful test.

Answer. Delete the midpoint of $[0,1]$. The two resulting half-intervals give a separation. Deleting any point from the circle leaves a space homeomorphic to $\mathbb{R}$. Rotate that point to the north pole, then use Lecture 1's stereographic projection. The line is path-connected by straight segments. A homeomorphism would restrict to a homeomorphism between these two punctured spaces, which is impossible. Deleting an endpoint leaves a path-connected half-open interval, so connectedness would not distinguish them.

Summary. Continuous images preserve compactness and connectedness. A continuous bijection from compact to Hausdorff is a homeomorphism. A quotient topology tests continuity after composition with its projection. Quotients preserve compactness and path-connectedness, but Hausdorffness needs its own proof. The rotation group is the closed three-ball with antipodal boundary points identified, hence $\mathbb{R}P^3$. Lecture 3 begins the study of loops.

Reading. Munkres, *Topology*, second edition, §§22–28; Heine–Borel is Theorem 27.3, metric sequential compactness is Theorem 28.2, and the sine curve is §24, Example 7.